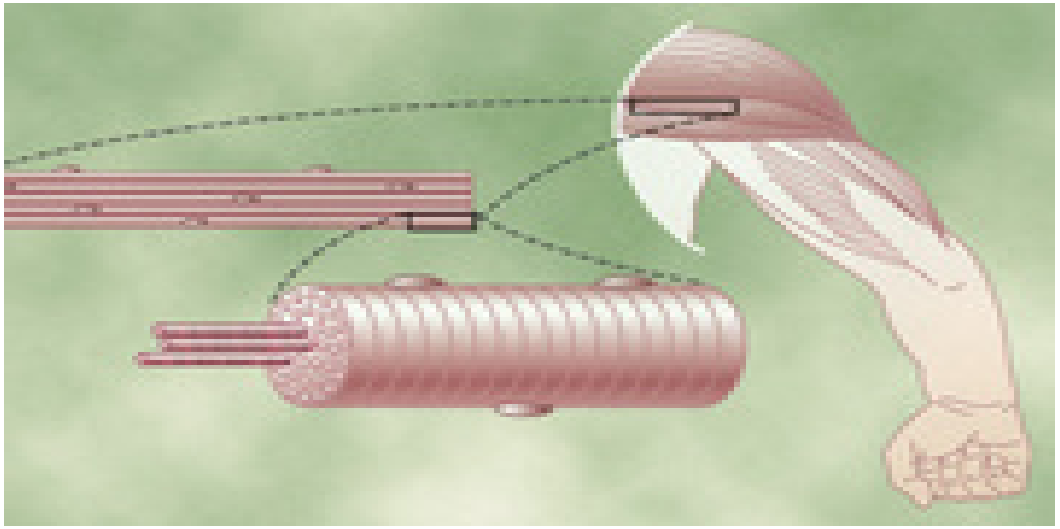


Physiological Anatomy of Skeletal Muscle and Contraction mechanism

Dr.Sidra Zafar (Lecturer,FUUAST)

Edit with WPS Office

Introduction

- Skeletal muscle is composed of thousands of individual muscle f bers, which range in diameter from 10 to 80 micrometers. Each of these f bers is itself constructed from progressively smaller structural subunits.
- Nearly all skeletal muscle f bers span the entire length of the muscle. Almost every f ber (about 98%) is controlled by just one nerve connection, and this connection is located close to the middle of the fiber.

Edit with WPS Office

Properties of Muscle

Four main Properties and characteristics

1. Excitability: Responds to neurotransmitters (ACh) → generates action potentials.

- Resting membrane potential: -80 to -90 mV in skeletal muscle
- Action potential duration: 1–5 min, conduction velocity: 3–5 m/s
- All-or-none law applies to single muscle fibers

2. Contractility: Shortens actively to produce force.

- Maximum force: 3–4 kg/cm² cross-sectional area
- Summation & tetanus occur at frequencies >30 Hz

3. Extensibility: Can stretch without damage

4. Elasticity: Returns to original length after stretch.

- Parallel elastic component (connective tissue)
- Series elastic component (tendons, cross-bridges)

EN STUDYNOTES.COM

Major Functions

- **Movement Production:** Shortening of muscle fibers pulls on tendons to move the skeleton.
- **Posture Maintenance:** Continuous, low-level contractions resist gravity to keep the body upright.
- **Joint Stabilization:** Muscle tone and co-contraction around joints provide stability during movement.
- **Heat Generation:** Inefficiency of ATP use during contraction releases thermal energy (shivering thermogenesis).

Edit with WPS Office

Skeletal Muscle

- **Introduction**
- **Striated Appearance:** Due to highly organized sarcomeres composed of actin and myosin filaments arranged in repeating A-bands and I-bands.
- **Voluntary Control:** Innervated by somatic motor neurons via the neuromuscular junction (NMJ), where acetylcholine (ACh) initiates excitation-contraction coupling.
- **Multinucleation:** Each fiber arises from the fusion of myoblasts, resulting in multiple peripheral nuclei—a feature essential for protein synthesis and repair.

Edit with WPS Office

Muscle Attachment & Connective Tissue

- **Muscle–Bone Connection: Tendon**
- **Tendon: Dense Regular Connective Tissue:** Parallel bundles of **Type I collagen** (≥90% dry weight) secreted by fibroblasts (**tenocytes**).

1. **Structure:** Contains **crimped** collagen f brils for elasticity; collagen bundles are surrounded by endotenon and entire tendon is encased in paratenon or synovial sheath for gliding.
2. **Biomechanical Role: Tendon Stiffness** and **Strain Energy Storage**. Acts as a biological spring to enhance movement efficiency.
3. **Attachment via Fibrocartilaginous Enthesis:** At the bone interface, tendon transitions through four distinct zones: tendon → f brocartilage → mineralized f brocartilage → bone. This gradation dissipates stress and prevents failure.

Edit with WPS Office

Muscle Attachment & Connective Tissue

Muscles connect to bones like two ways to tie a rope to a pole

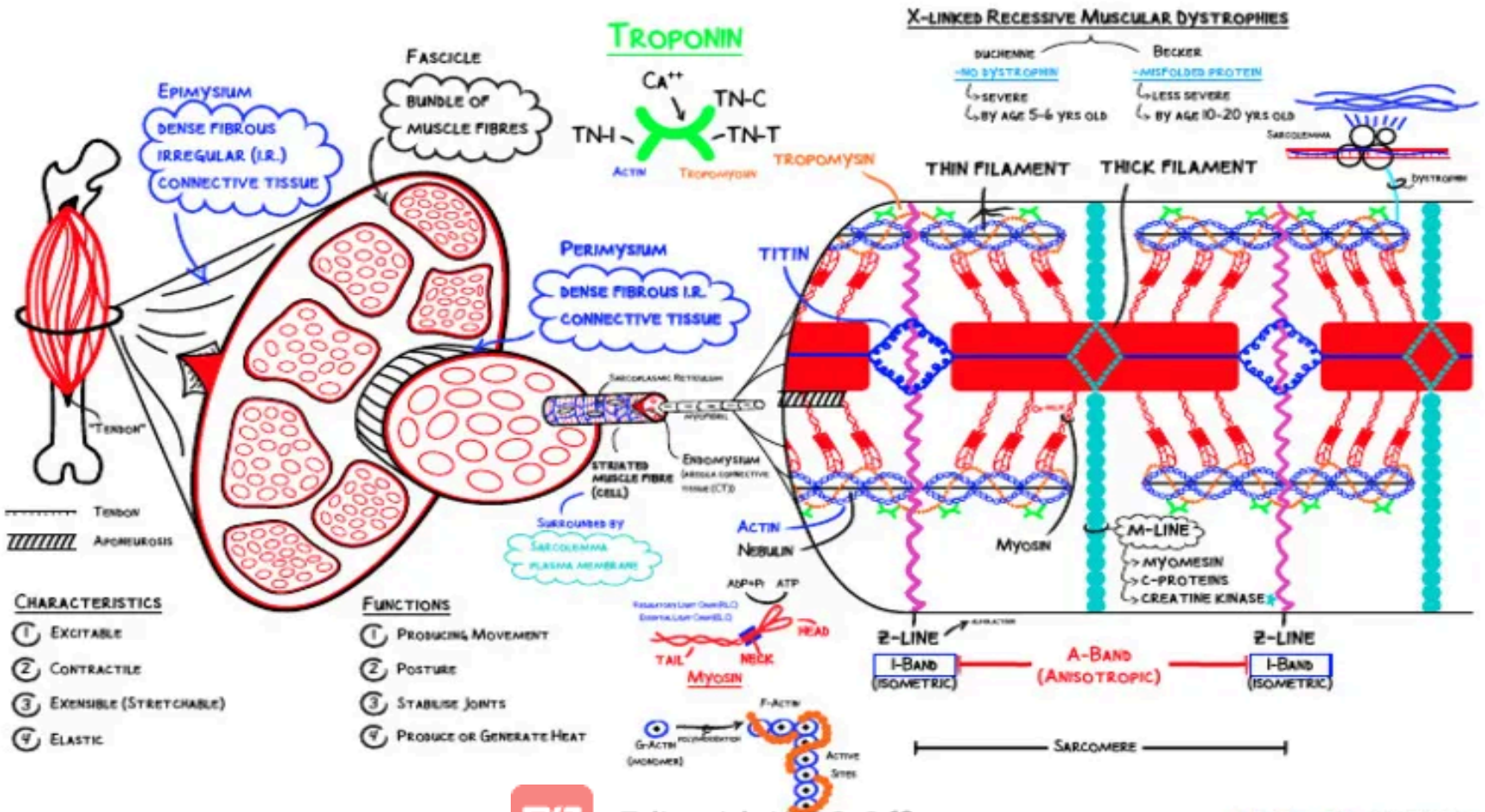
Feature	Direct Attachment (Fleshy)	Indirect Attachment (Common)
What it is	Muscle tissue **glues directly** to bone	Muscle connects via a **rope-like tendon**
How it looks	Muscle → Bone (no rope)	Muscle → Tendon → Bone
Common?	Rare (few places)	Very common (most muscles)
Like this	Velcro strip on a wall	Jump rope tied to a fence
Strength	Good for spreading force over area	Good for leverage and spring action
Example	Trapezius muscle on shoulder blade	Biceps muscle (has a tendon to elbow bone)
Injury risk	Can tear muscle off bone	Can snap the "rope" (tendon)

Edit with WPS Office

Muscle Attachment & Connective Tissue

- **Connective Tissue Layers:**
  - **Epimysium:** Surrounds entire muscle.
  - **Perimysium:** Surrounds fascicles.
  - **Endomysium:** Surrounds individual muscle fibers.

Edit with WPS Office



Sarcomere Structure: Actin and Myosin

Edit with WPS Office

NINJA NERD

Structural Components of a Skeletal Muscle Fiber

1. The Sarcolemma: The Fiber's Outer Membrane

- The specialized cell membrane enclosing a muscle fiber.
- **Two-Layer Structure:**
  - **Plasma Membrane:** The inner, true cell membrane.
  - **Outer Basement Membrane:** A polysaccharide coat containing collagen fibrils.
- **Function:** At each end, the sarcolemma fuses with **tendon fibers**, which bundle together to form the muscle tendons that attach to bone.

Edit with WPS Office

Structural Components of a Skeletal Muscle Fiber

2. The Contractile Elements: Myofibrils and Filaments

- **Myof brils:** Hundreds to thousands of rod-like structures packed inside each f ber. They are the contractile machinery.
- **Filament Composition (Per Myofibril):**
  - **~1,500 Myosin Filaments** (Thick)
  - **~3,000 Actin Filaments** (Thin)

Edit with WPS Office

Structural Components of a Skeletal Muscle Fiber

3. The Striated Pattern: A Bands and I Bands

- The interdigitation of actin and myosin creates alternating light and dark bands.
- **I Band (Light Band):**
  - Contains **actin filaments only**.
  - **Isotropic** to polarized light.
- **A Band (Dark Band):**
  - Contains the full length of the **myosin filaments**.
  - Includes the overlapping ends of actin filaments.
  - **Anisotropic** to polarized light.

Edited with WPS Office

Structural Components of a Skeletal Muscle Fiber

4. The Sarcomere: The Basic Contractile Unit

- The segment of a myofibril between two successive **Z Disks**.
- **Z Disk:**
  - A protein lattice to which actin filaments are anchored.
  - Links adjacent myofibrils together across the entire fiber, transmitting force.
- **Sarcomere Length (Contracted):** ~2.0 μm. At this length, actin filaments fully overlap myosin, enabling maximal force generation.

Edit with WPS Office

Structural Components of a Skeletal Muscle Fiber

5. The Molecular Spring: Titin

- **Function:** Giant, elastic filamentous protein that maintains myofilament alignment.
- **Role:**
  - **Structural Framework:** Holds myosin and actin in their precise positions.
  - **Elastic Spring:** Attaches from the Z disk to the myosin filament. Stretches and recoils during contraction/relaxation.
  - **Template:** Aids in the initial assembly of the sarcomere.

Edit with WPS Office

Structural Components of a Skeletal Muscle Fiber

6. The Internal Environment: Sarcoplasm

- The intracellular fluid that fills the spaces between myofibrils.
- **Components:**
  - **Ions:** High [K^+], [Mg^{2+}], phosphate.
  - **Enzymes:** Numerous protein enzymes for metabolism.
  - **Mitochondria:** Lie parallel to myofibrils.
    - **Critical Role:** Supply **ATP** for contraction.

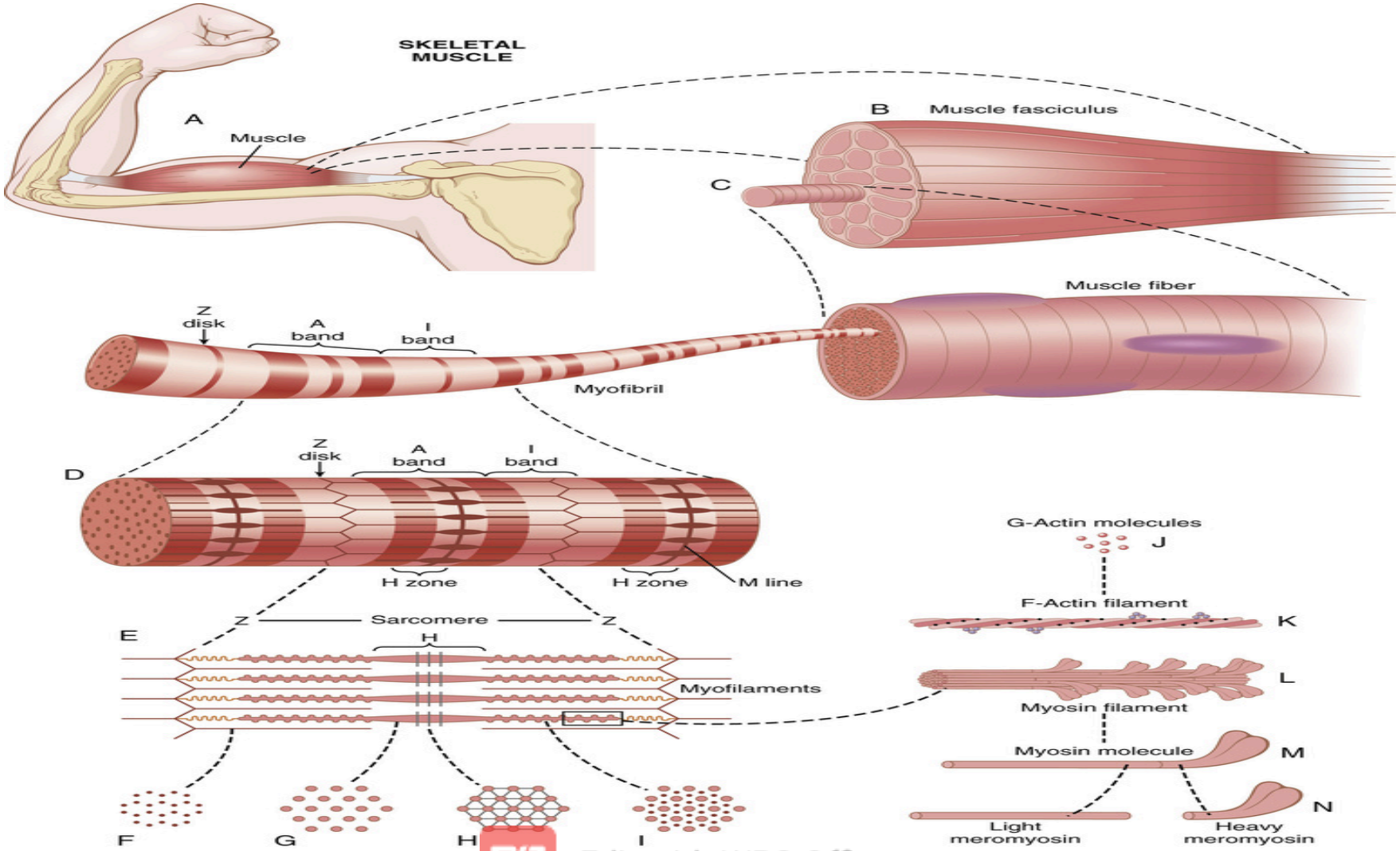
Edit with WPS Office

Structural Components of a Skeletal Muscle Fiber

7. The Calcium Reservoir: Sarcoplasmic Reticulum (SR)

- **Definition:** A specialized endoplasmic reticulum that forms a network around each myofibril.
- **Primary Function: Regulates intracellular calcium.**
  - **Stores Ca^{2+}** at rest.
  - **Releases Ca^{2+}** to trigger contraction.
  - **Rapidly re-accumulates Ca^{2+}** to cause relaxation.
- **Specialization:** Fast-twitch muscle fibers have a more extensive SR for rapid calcium cycling.

Edit with WPS Office



Edit with WPS Office

What happens in the body that leads to muscle growth after exercise?

Edit with WPS Office

Thank You